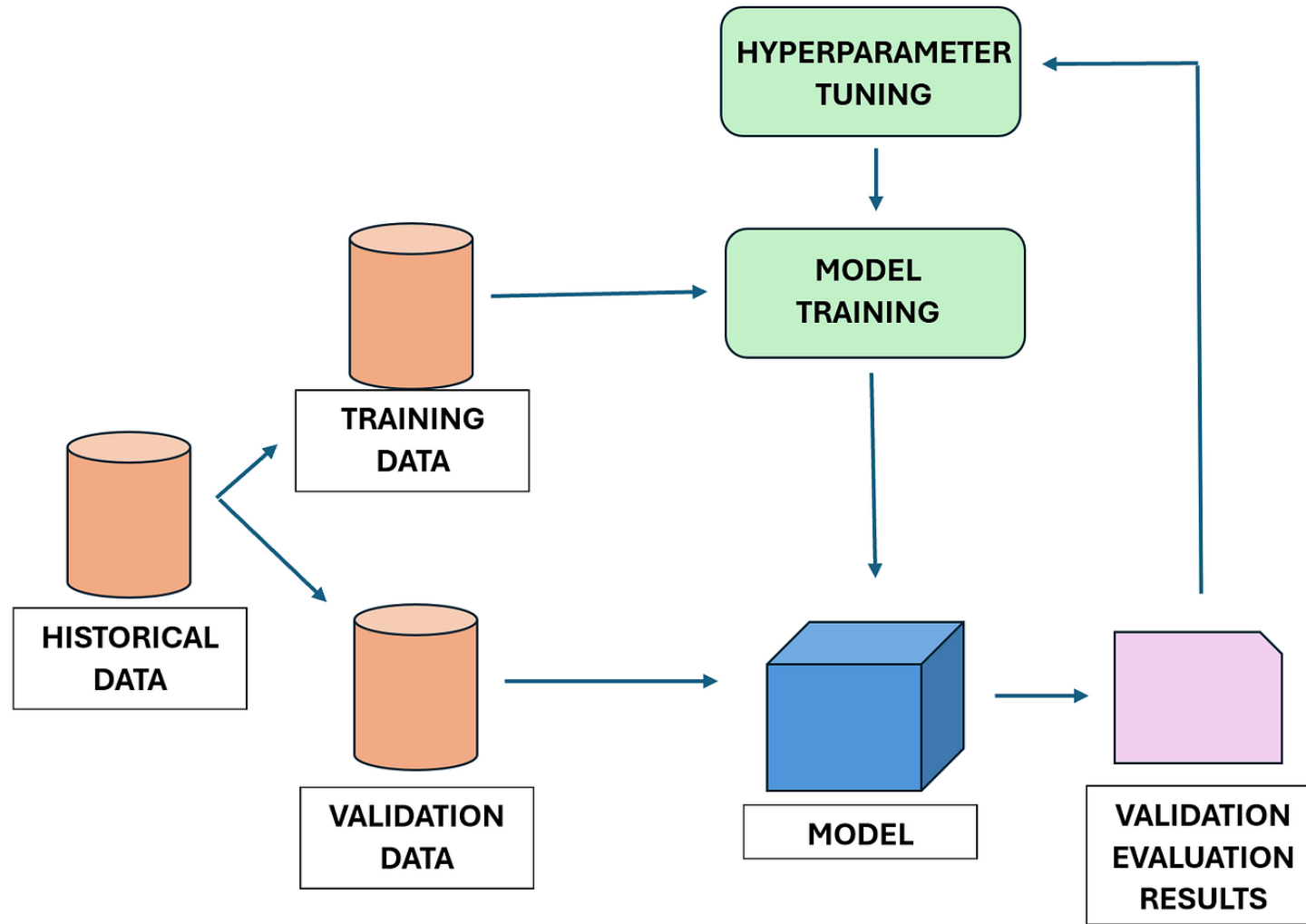


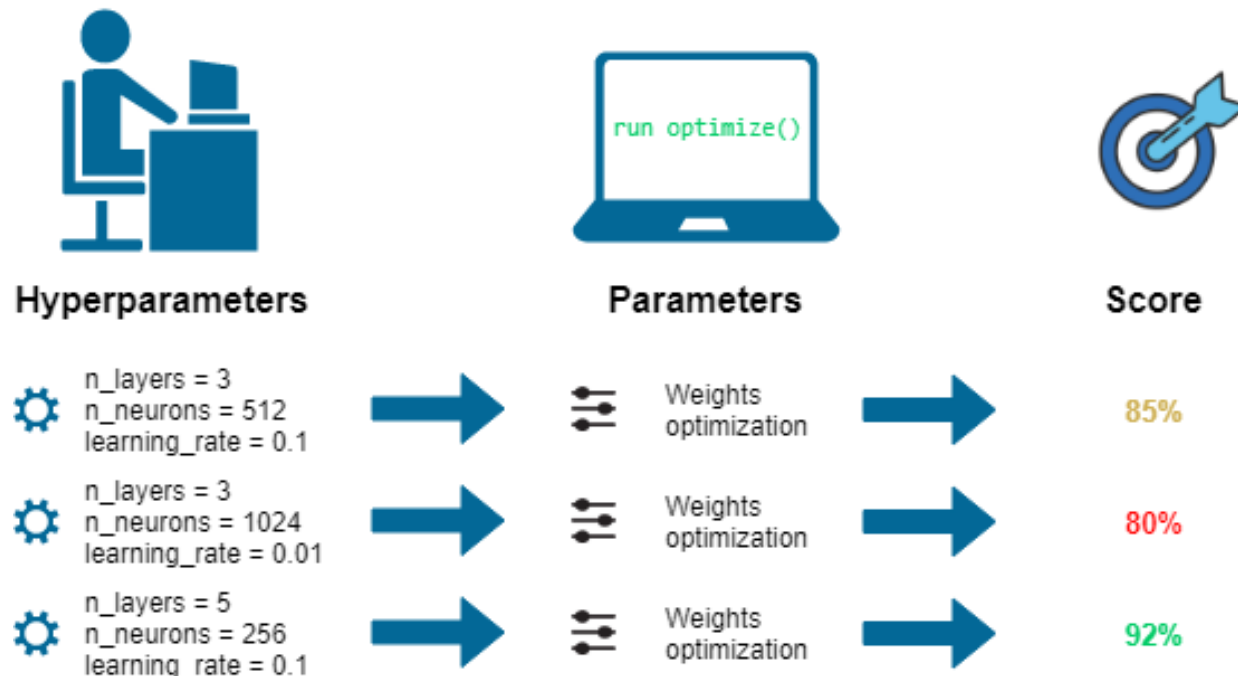
IMPROVING THE MODEL (HYPERPARAMETER TUNING)



Turning the Dials for Peak Performance

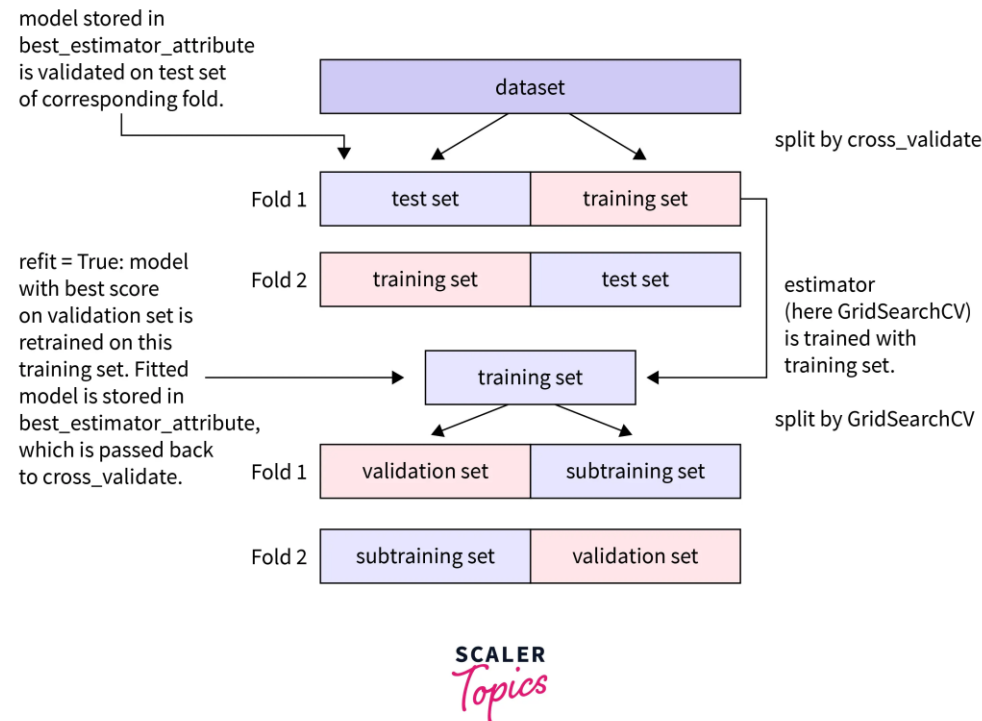
Instructor: Zaryab Ahmad Khan

WHAT ARE HYPERPARAMETERS?



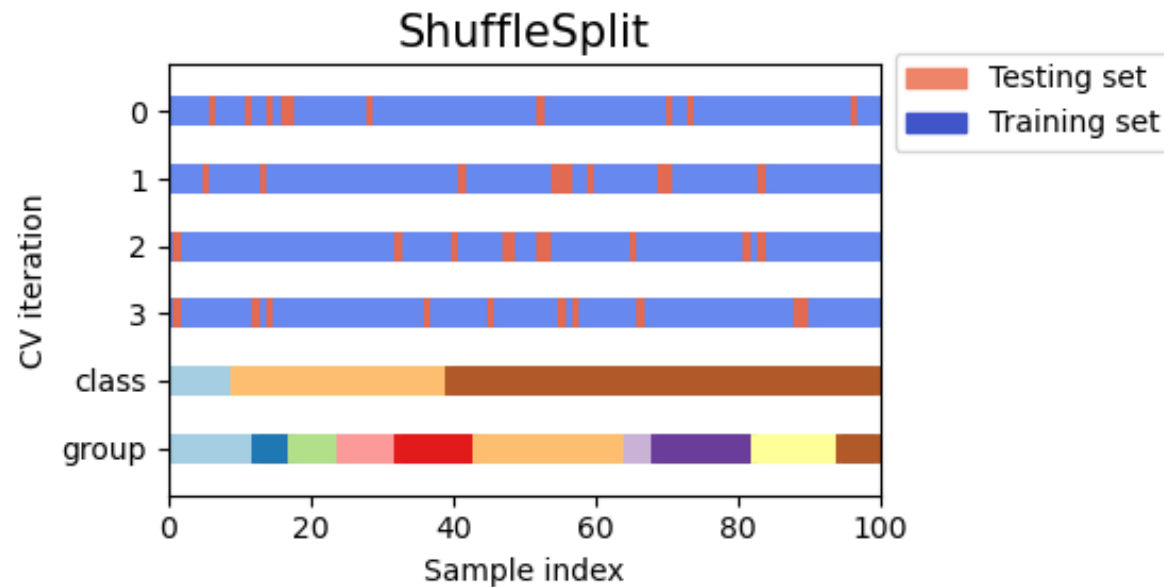
- **The Concept:** These are the "settings dials" on an algorithm.
- **The Difference:** Unlike regular "parameters" (which the AI learns on its own), "hyperparameters" are the rules **you** set before training starts.
- **Example:** Telling a Random Forest exactly how many trees to build (e.g., 50 vs. 500).

GRID SEARCH (GRIDSEARCHCV)



- **The Problem:** How do you know if 100 trees or 200 trees is better? Testing one by one is slow.
- **The Solution:** Grid Search.
- **Process:** You give the computer a list of different settings, and it forces the AI to test every single possible combination to find the most accurate setup.

CROSS-VALIDATION (CV)



- **The Goal:** Reliability.
 - **The Method:** Instead of testing the model just once, we slice the data into 5 different parts (5-fold CV).
 - **The Benefit:** The model is tested 5 different times on different slices of the data. If it performs well on all of them, you know your results are reliable and not just a "lucky" split.
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